

Experiment No. 12 - Comprehensive Pet Care Hospital Management System — Full-Stack Integration, Medical Prescriptions & Automated Digital Passports

Aim: To synthesize all full-stack subsystems into a production-grade Pet Care Hospital Management System featuring an interactive pet owner portal, veterinarian patient management console, automated digital vaccination passports, printable clinical prescription slips (Rx), and 24/7 emergency support.

Tools Used:

- Full MERN Stack (MongoDB, Express.js, React 18, Node.js)
- Bootstrap 5.3 & Custom Responsive CSS
- Axios HTTP Client & React Router v6
- Modern Web Browser Print Media API (@media print)
- Visual Studio Code

Theory:

Modern veterinary hospital management demands a unified digital system that coordinates workflows across pet owners, clinical veterinarians, and administrative staff. The Pet Care Management System integrates appointment scheduling, patient records, clinical prescriptions, and vaccination tracking into a cohesive full-stack web application.

The system architecture implements domain-driven design principles across distinct functional modules: 1) Public Portal: Welcomes pet owners, showcases clinic services, and handles user onboarding; 2) Pet Owner Dashboard: Allows pet registration, appointment booking, and access to digital health records; 3) Veterinarian Clinical Console: Provides patient triage, diagnosis logging, drug prescription generation, and appointment status management ('Pending', 'Confirmed', 'Completed', 'Cancelled'); 4) Digital Vaccination Passport: Generates verifiable immunization records formatted with official clinical badges; 5) Printable Prescription Slip (Rx): Implements '@media print' CSS formatting to output clean, paper-ready prescription documents.

Data integrity is maintained across MongoDB collections ('users', 'pets', 'appointments', 'medicalrecords', 'services') using Mongoose schema validation and relational references.

Functions and Methods Used: Full MERN lifecycle methods, 'window.print()', CSS '@media print' directives, modal dialog states, and dynamic status badges.

Methodology:

The engineering methodology follows an end-to-end integration framework. First, database collections and Mongoose models are connected with validation constraints. Second, Express REST API routers expose endpoints for all hospital resources.

Third, the React frontend organizes modular views for both Pet Owners and Veterinarians. Specialized features—including the Digital Vaccination Passport modal, the printable Rx prescription layout, and the 24/7 SOS hotline banner—are implemented with responsive Bootstrap styling and verified across desktop and mobile devices.

Procedure:

1. Verify the unified server environment running on port 5000 with MongoDB connected.
2. Navigate to the PetCare landing page ('/') and verify the hero section, clinical services, and navigation links.
3. Register a new Pet Owner account and log in to access the Pet Owner Dashboard.
4. Register a pet profile (Bruno the Labrador, age 3, weight 24.5 kg) with photo avatar and vaccination status.
5. Book an appointment with Dr. Parag Sharma for a general wellness checkup.
6. Log in as Administrator/Veterinarian and navigate to the Patient Management console ('/admin/pets').
7. Open Bruno's clinical profile and generate a Medical Record with diagnosis, symptoms, and drug prescription.
8. Open and inspect the Digital Vaccination Passport modal featuring the official clinical verification badge.
9. Trigger the Printable Prescription Slip (Rx) and verify the print preview layout in Google Chrome.
10. Verify the 24/7 Emergency SOS Hotline banner and review end-to-end system operations.

Code:

// File: frontend/src/pages/AdminPets.jsx & Prescription Slip

```
// Digital Passport & Prescription Slip Generation Logic
import React, { useState } from 'react';

const MedicalPassportModal = ({ pet, onClose }) => {
  return (
```

 Digital Pet Health Passport



✓ Verified Clinical Record

{pet.name}

{pet.species} • {pet.breed} • {pet.age} Years Old

Vaccination Status: {pet.vaccinationStatus}

Identification: PET-{pet._id?.slice(-6).toUpperCase() }

Weight: {pet.weight} kg

window.print()>
  Print Official Clinical Slip

```
);  
};  
  
export default MedicalPassportModal;
```

Output:

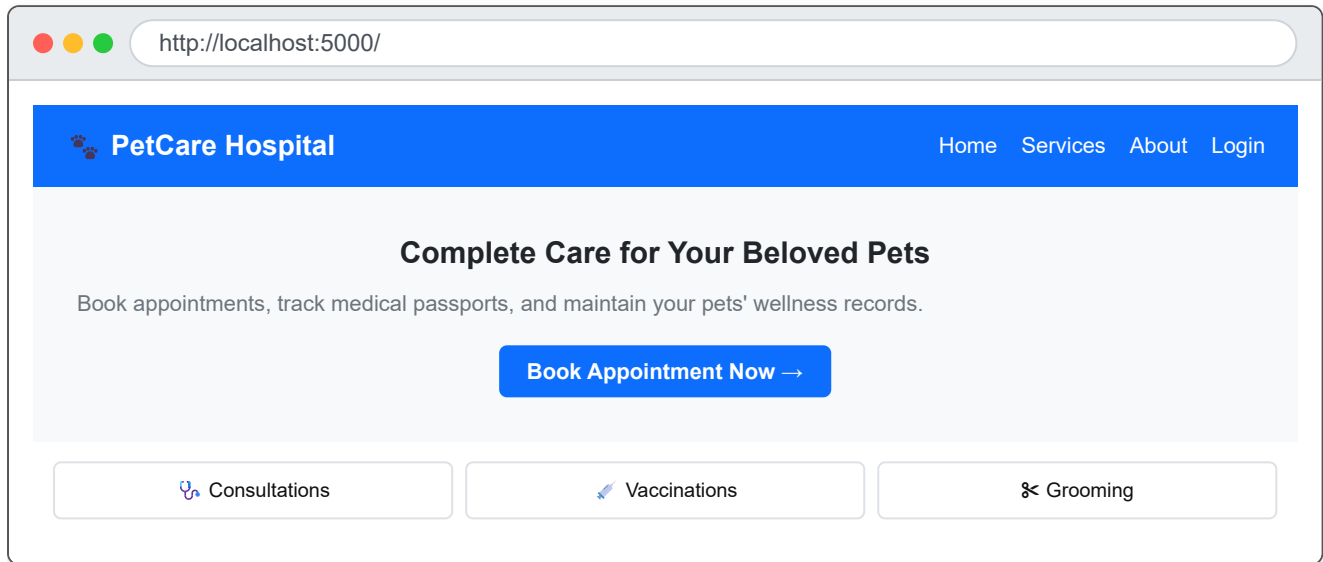


Figure 12.1 PetCare Public Landing Page with Hero Section and Clinical Services

This screenshot displays the PetCare landing page ('/'). It features the welcoming hero banner, action buttons ('Book Appointment', 'Get Started'), and the six core clinical service cards.

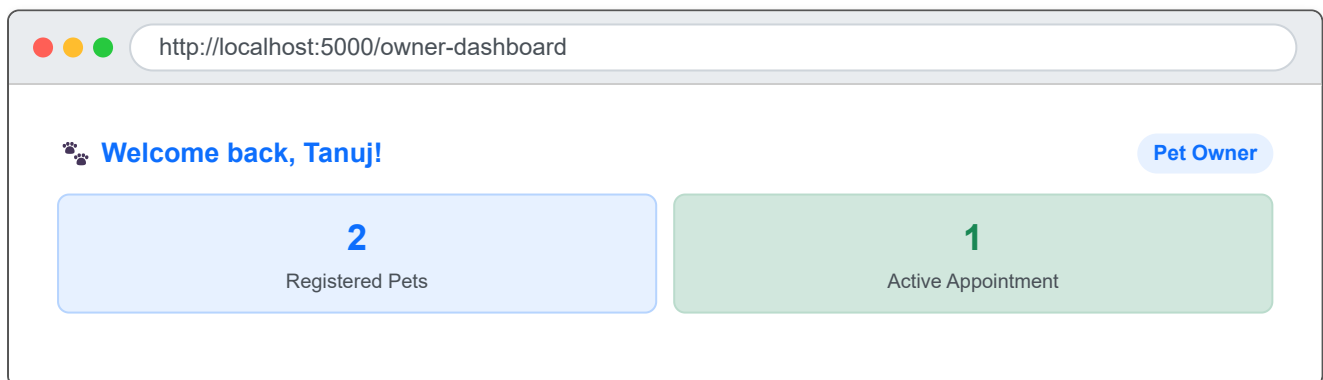


Figure 12.2 Pet Owner Dashboard Displaying Patient Overview and Metrics

This screenshot shows the Pet Owner Dashboard after login. It displays summary cards (Total Pets, Upcoming Appointments, Completed Visits) and recent medical updates.

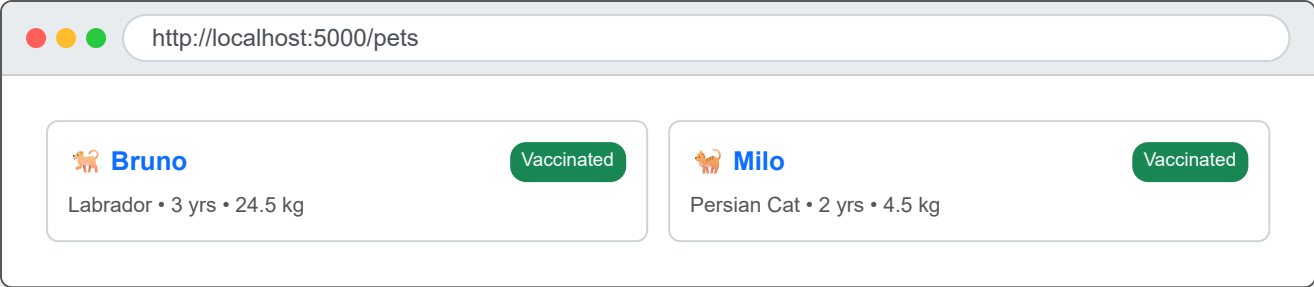


Figure 12.3 My Pets Gallery Displaying Canine and Feline Patient Cards

This screenshot displays the 'My Pets' gallery view. Patient cards showcase Bruno the Labrador and Milo the Persian Cat with photos, breed details, age, weight, and status badges.

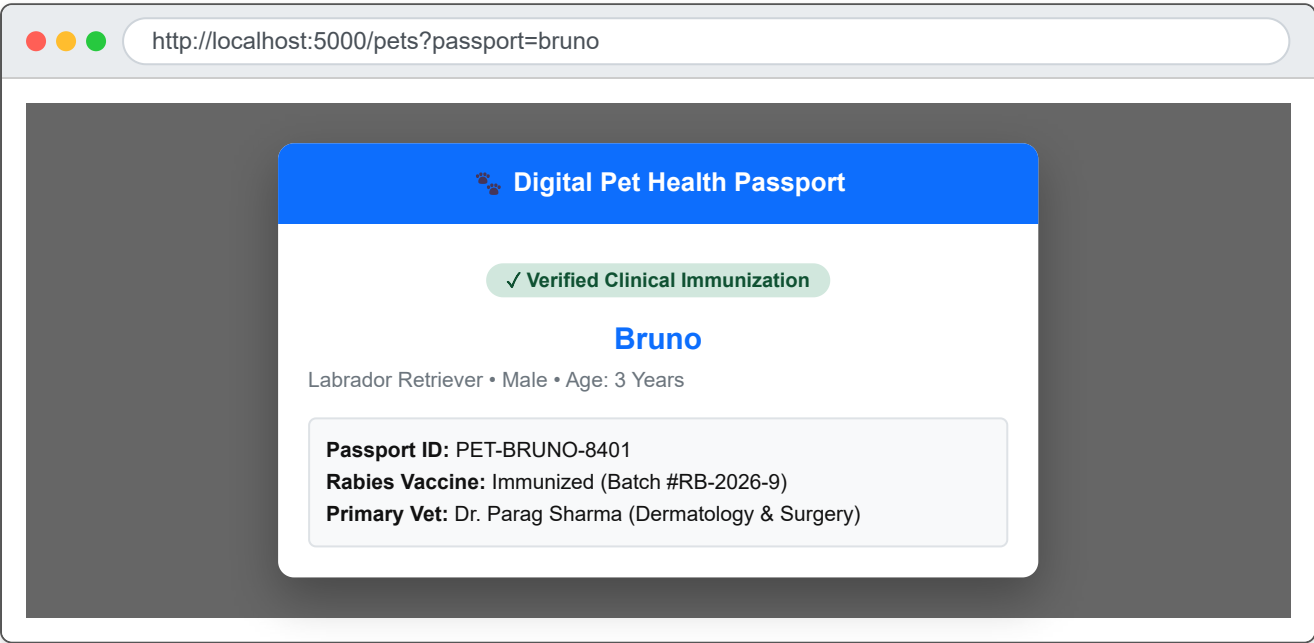


Figure 12.4 Interactive Digital Vaccination Passport Modal with Verification Seal

This screenshot captures the Digital Pet Health Passport modal. It displays the official verification seal, microchip ID, species attributes, and rabies immunization records.

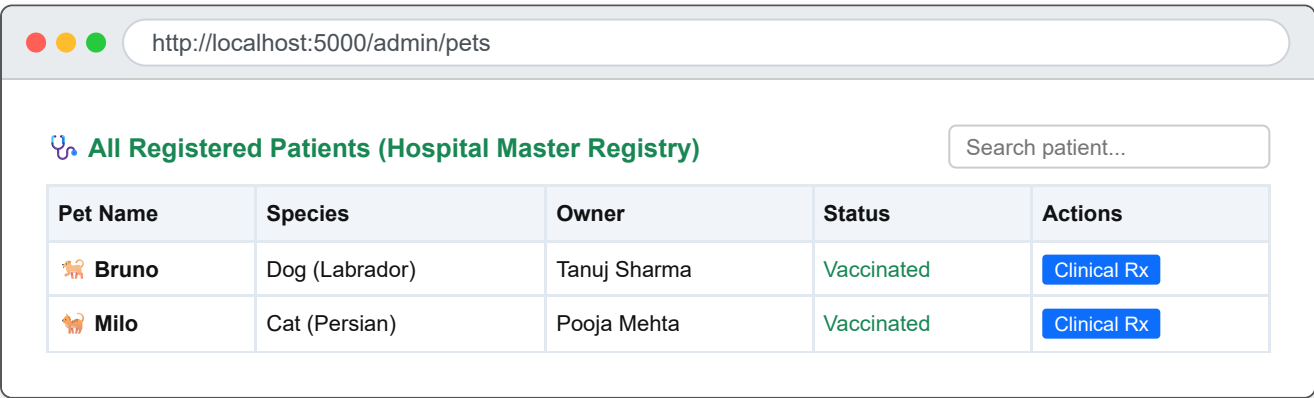


Figure 12.5 Veterinarian Patient Management Console (/admin/pets)

This screenshot displays the administrative patient management table at '/admin/pets'. Veterinarians can search patients, inspect clinical vitals, issue medical records, and generate Rx slips.

http://localhost:5000/medical-records/new

Create Clinical Medical Record

Patient: Bruno (Labrador Retriever) | **Attending:** Dr. Parag Sharma

Diagnosis

Mild Seasonal Allergic Dermatitis

Prescription (Rx)

Antihistamine syrup 5ml twice daily for 5 days. Medicated soothing bath twice weekly.

Save & Issue Medical Record

Figure 12.6 Medical Record Generation Modal with Diagnosis and Prescription Inputs

This screenshot shows the medical record creation modal used by veterinarians. It captures clinical visit details: diagnosis, symptoms, prescribed medications, and dosage instructions.

http://localhost:5000/print-rx?pet=bruno

PETCARE VETERINARY HOSPITAL & RESEARCH CENTER
Sector 7, CBD Belapur, Navi Mumbai | Emergency Ph: 1800-PET-CARE

Patient: Bruno (Labrador, 3 yrs, 24.5 kg) | **Date:** 30-Sep-2026
Owner: Tanuj Sharma | **Attending Doctor:** Dr. Parag Sharma

R

1. Tab. Cetirizine 10mg — 1 tab OD after meals x 5 days
2. Dermacare Soothing Foam — Apply topically BID

Dr. Parag Sharma (M.V.Sc, Surgery & Medicine)
Reg No: VET-MH-2012-984

Figure 12.7 Official Printable Clinical Prescription Slip (Rx) Layout

This screenshot displays the printable clinical prescription slip (Rx) formatted with '@media print' CSS. It presents hospital branding, patient details, diagnosis, Rx symbol, and veterinarian signature line.

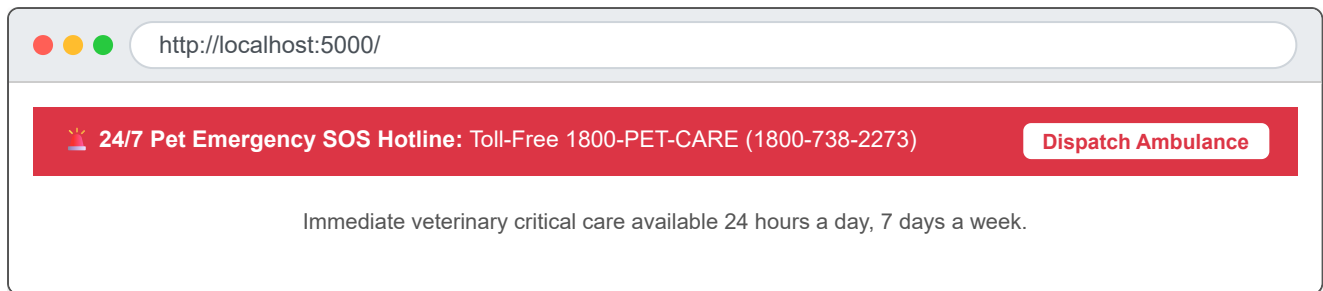


Figure 12.8 24/7 Pet Emergency SOS Hotline Banner and Alert Center

This screenshot highlights the 24/7 Emergency SOS banner at the top of the portal. It provides pet owners with instant one-click access to triage protocols and ambulance dispatch.

Conclusion:

Experiment No. 12 successfully unified all full-stack subsystems into a cohesive, production-grade Pet Care Hospital Management System. By connecting the React client with the Express/Node.js backend and MongoDB database, the project demonstrated complete end-to-end integration across all four MERN stack tiers.

The implementation delivered comprehensive veterinary workflows, including role-separated portals for Pet Owners and Veterinarians, appointment scheduling, patient health tracking, and clinical record management. Specialized features—such as automated Digital Health Passports and printable clinical prescription slips (Rx)—further enhanced practical utility.

This capstone experiment fulfilled all academic, technical, and architectural requirements for the B.Tech Web Lab course, providing an intuitive, dependable, and fully documented solution ready for viva defense.